

Mechanical Engineering

(ME10008)

Unit I

Mechanical Properties of Material

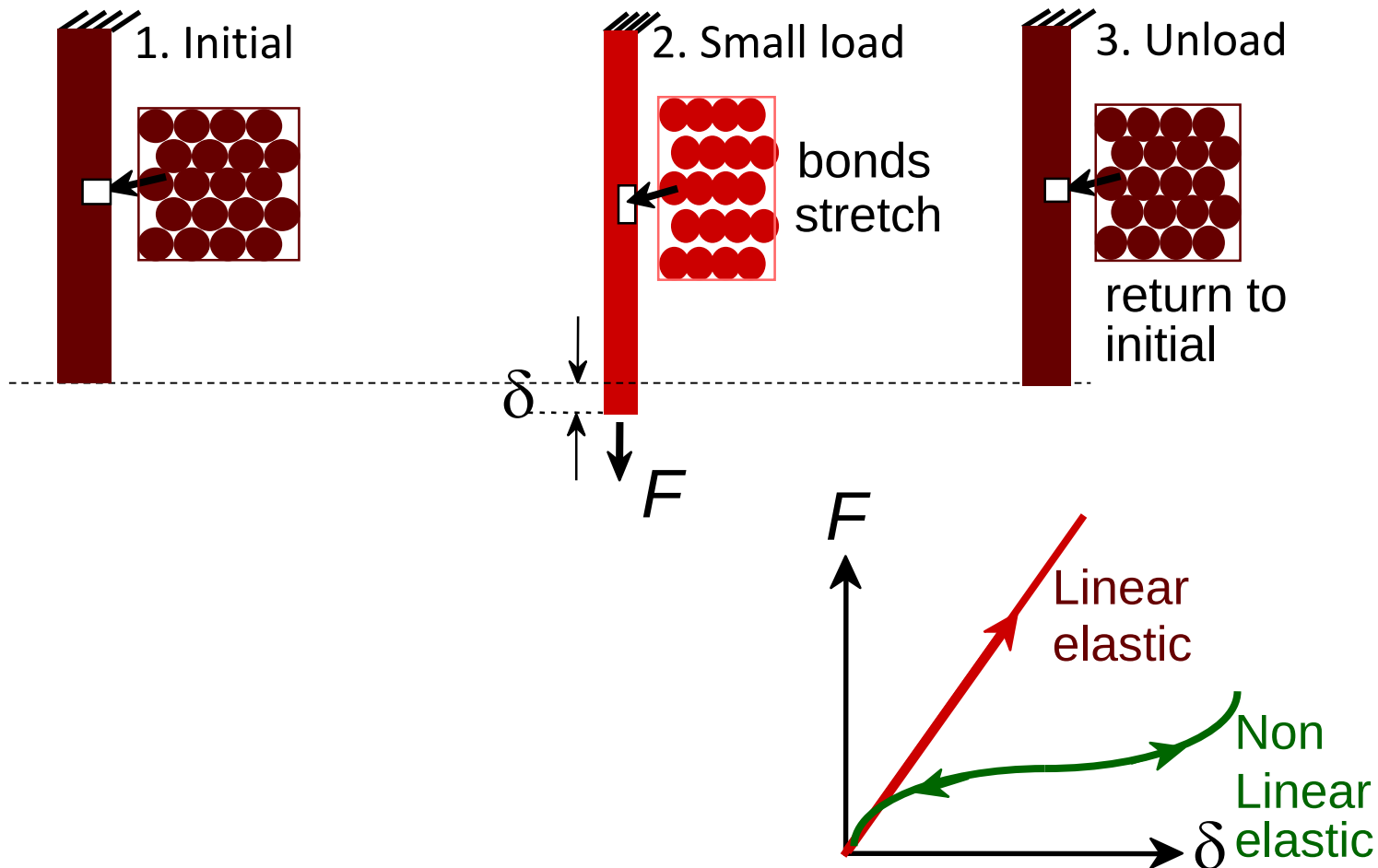
Mechanical Properties of Materials

- Elasticity
- Plasticity
- Ductility
- Brittleness
- Hardness
- Strength
- Malleability
- Toughness
- Resilience
- Impact Strength etc.

Elasticity or Elastic Deformation

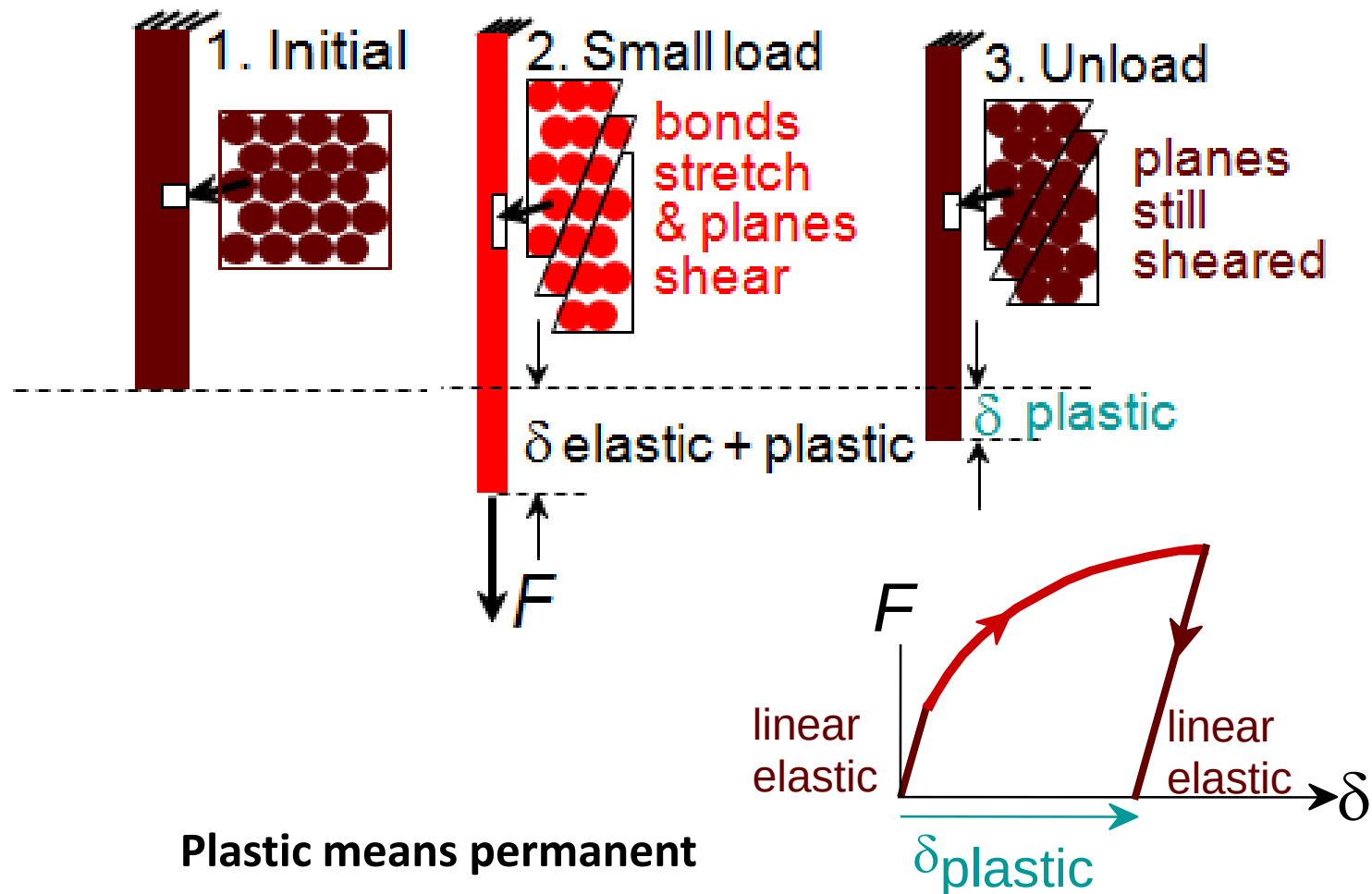
Elasticity is the ability of a material to resist a deformation influence and to return to its original size and shape when that influence or force is removed.

Solid objects will deform when adequate loads are applied to them; if the material is elastic, the object will return to its initial shape and size after removal of load.



Plasticity or Plastic Deformation

Plasticity is the ability of a solid material to undergo permanent deformation, a non-reversible change of shape in response to applied forces. Solid piece of metal being bent into a new shape displays plasticity as permanent changes occur within the material itself.

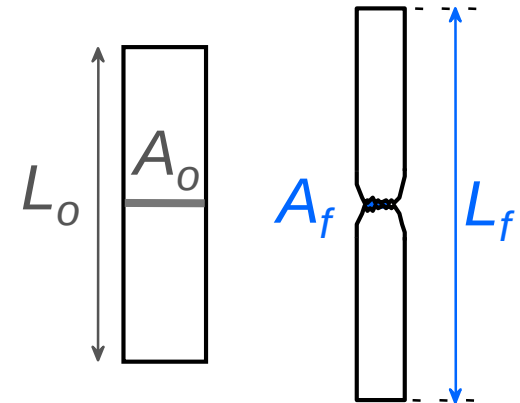
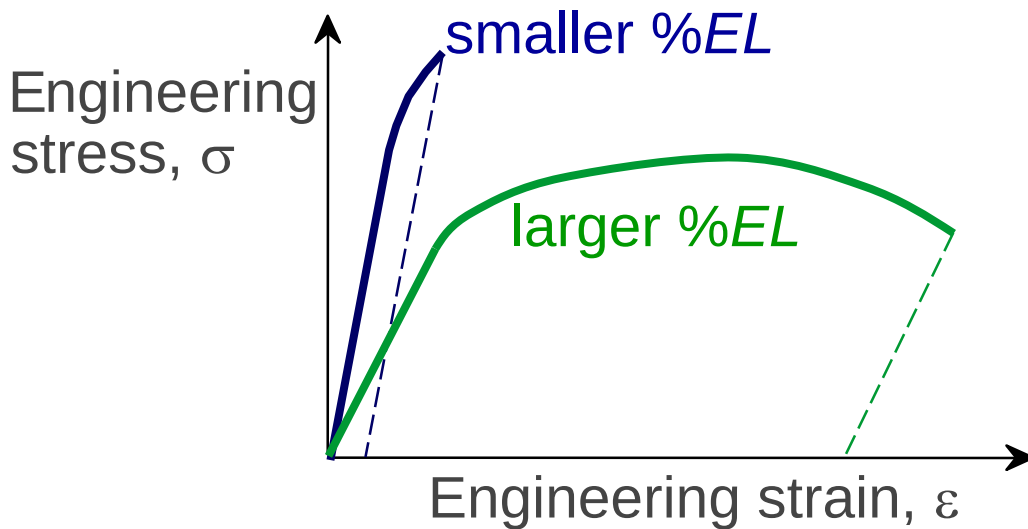


Ductility

Ductility is a measure of a material's ability to undergo significant plastic deformation before rupture or breaking, which may be expressed as percent elongation or percent area reduction from a tensile test.



Plastic tensile strain at failure:
$$\%EL = \frac{L_f - L_o}{L_o} \times 100$$



Another ductility measure:

$$\%RA = \frac{A_o - A_f}{A_o} \times 100$$



Cup-Cone Fracture

Brittleness

A material is brittle if, when subjected to stress, it breaks with little elastic deformation and without significant plastic deformation.



Brittle Fracture

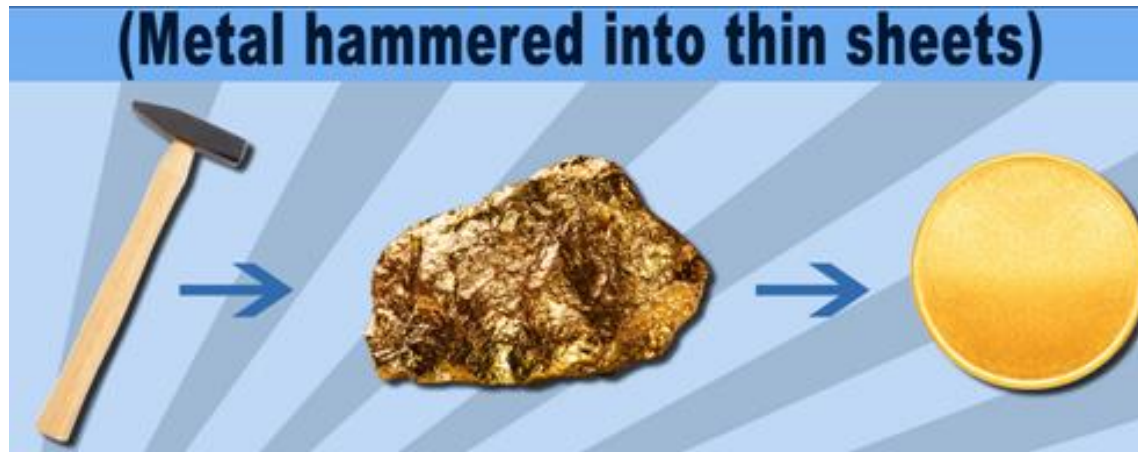
Strength

The **strength** of a material is its ability to withstand an applied load without failure or plastic deformation. The strength is the capacity to resist force or pressure.



Malleability

Malleability is a physical property of metals that defines their ability to be hammered, pressed, or rolled into thin sheets without breaking. In other words, it is the property of a metal to deform under compression and take on a new shape

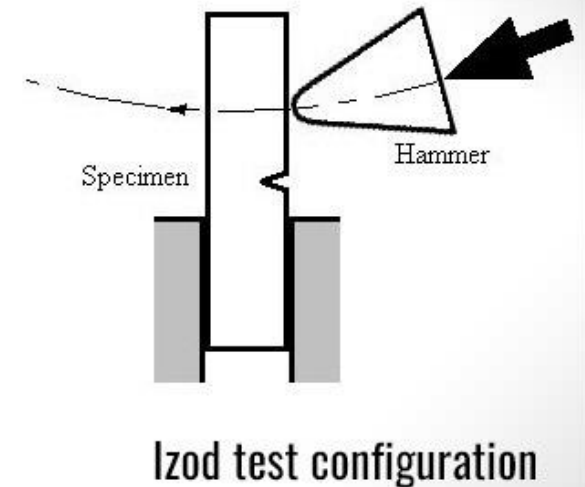
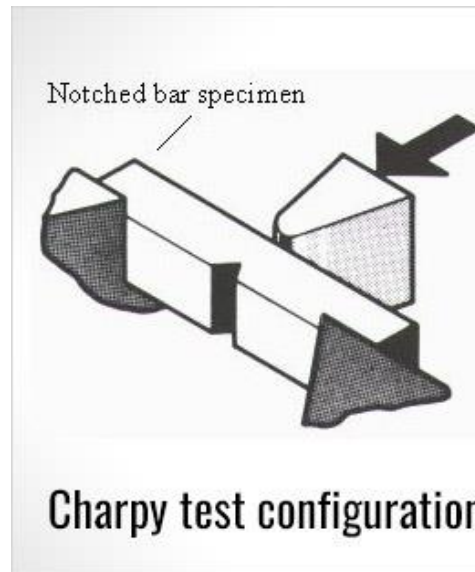
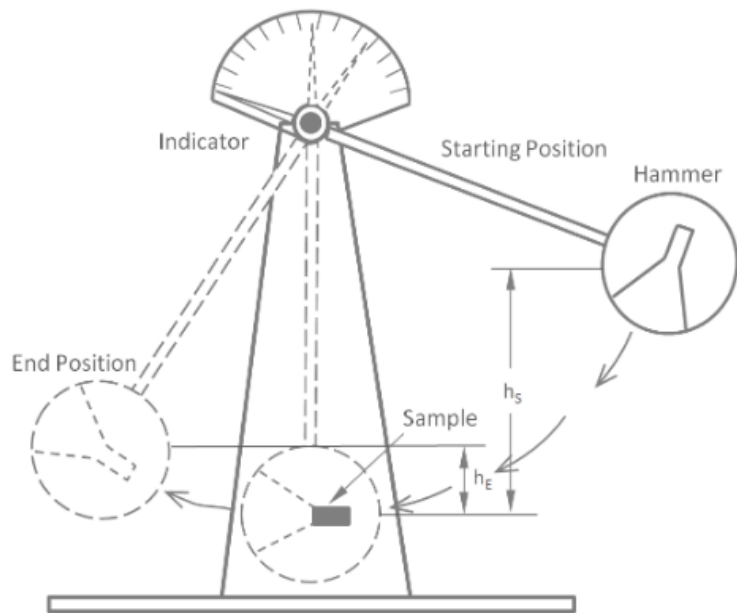


Gold is the most malleable metal.

Impact Strength

Impact strength is the capability of the material to withstand a suddenly applied load and is expressed in terms of energy.

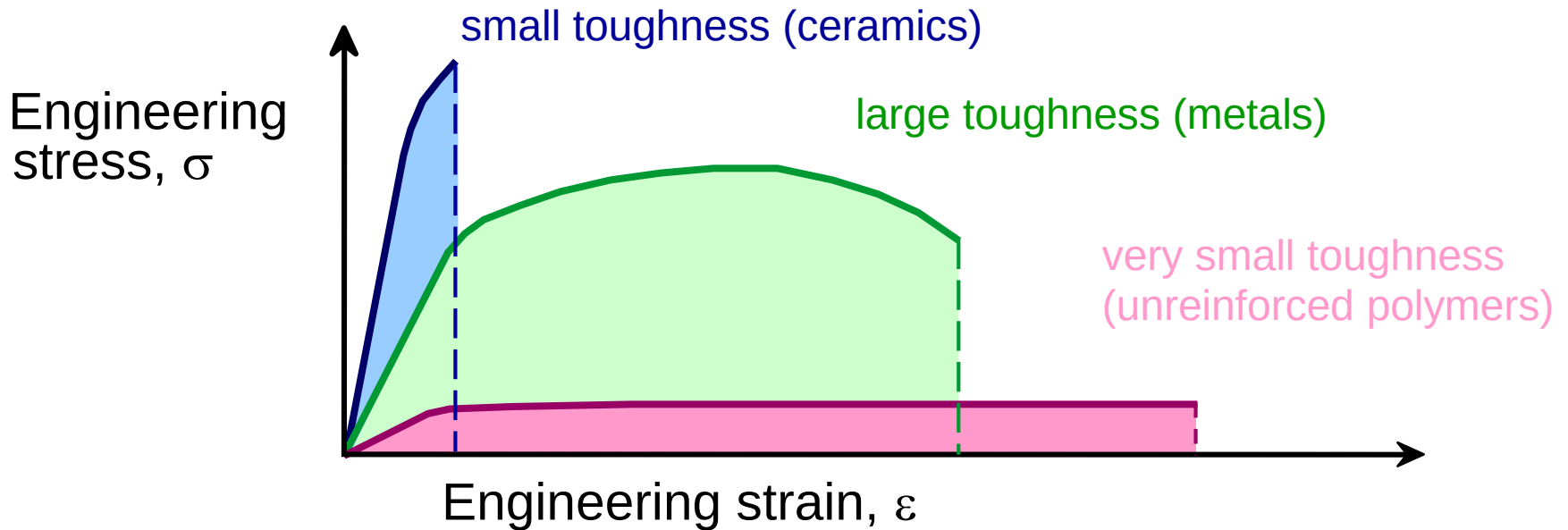
Often measured with the **Izod impact** strength test or **Charpy impact test**, both of which measure the impact energy required to fracture a sample



Impact Strength Test

Toughness

- Energy to break a unit volume of material
- Approximate by the area under the stress-strain curve.

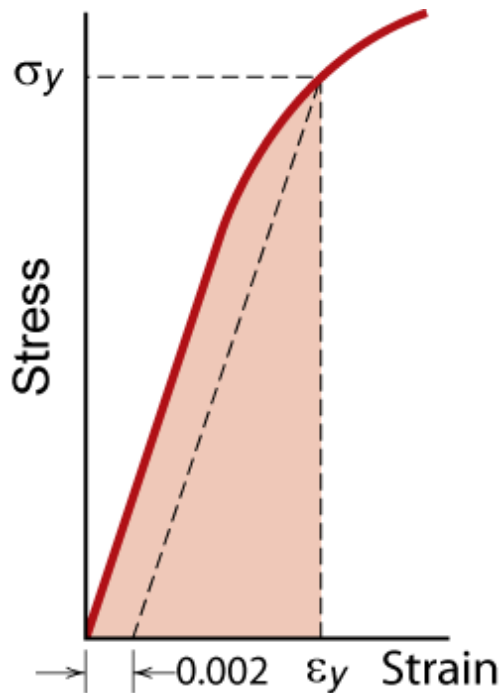


Brittle fracture: elastic energy

Ductile fracture: elastic + plastic energy

Resilience, U_r

- Ability of a material to store energy
 - Energy stored best in elastic region

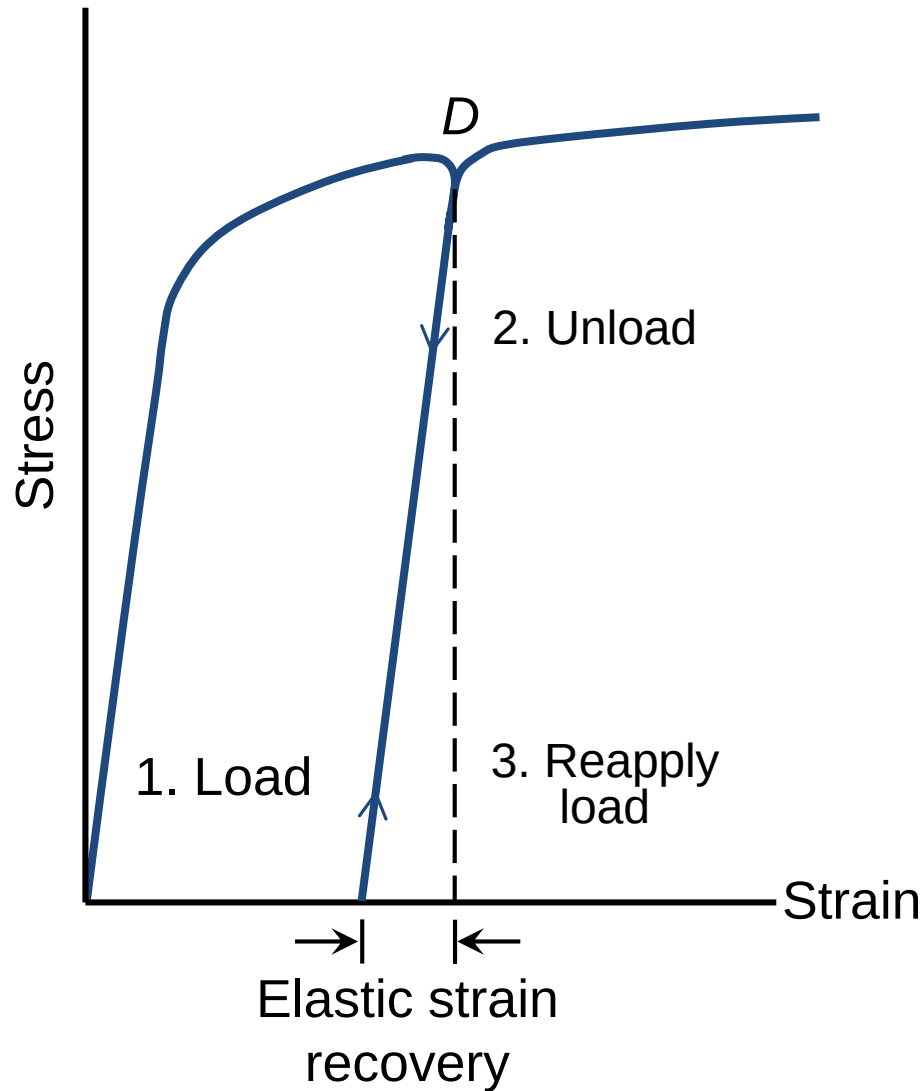


$$U_r = \int_0^{\epsilon_y} \sigma d\epsilon$$

If we assume a linear stress-strain curve this simplifies to

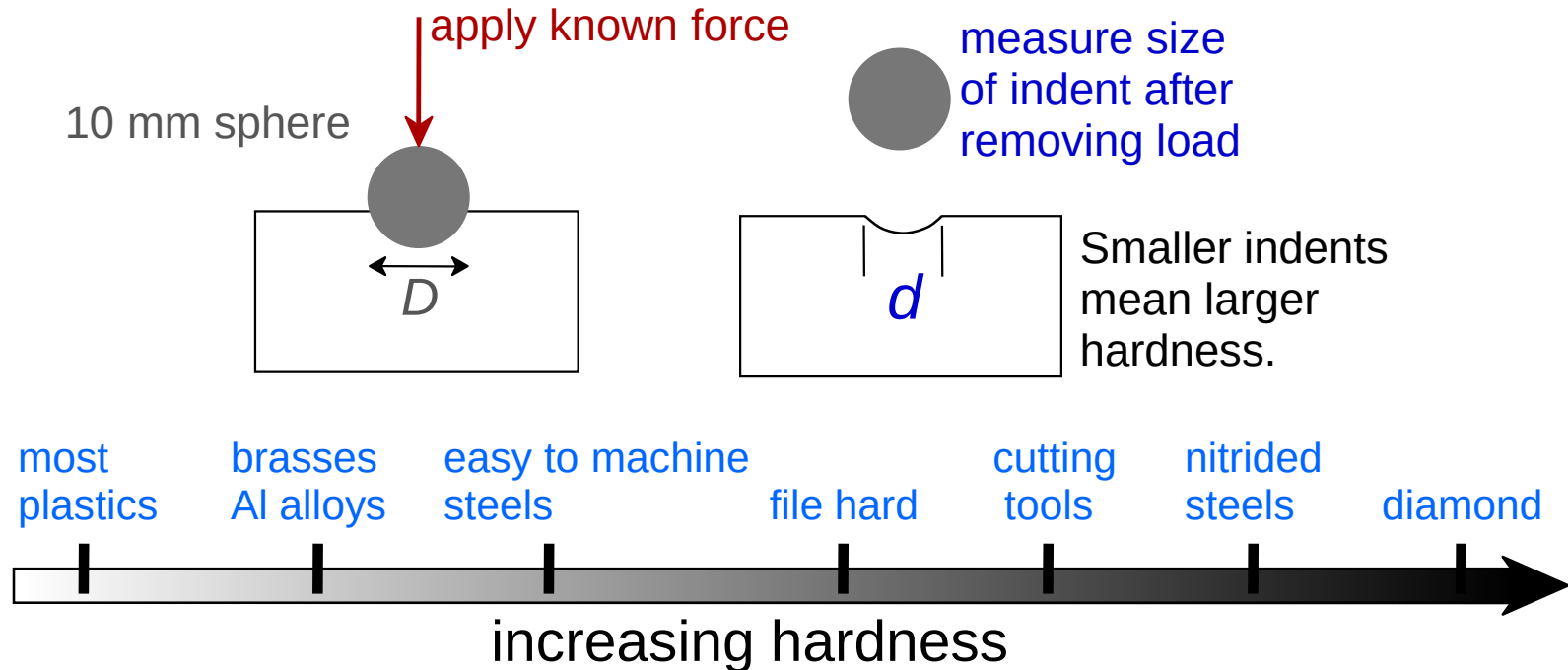
$$U_r \cong \frac{1}{2} \sigma_y \epsilon_y$$

Elastic Strain Recovery



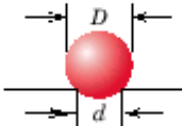
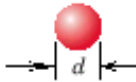
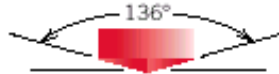
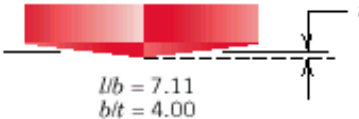
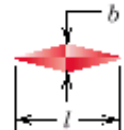
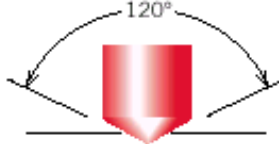
Hardness

- Resistance to permanently indenting the surface.
- Large hardness means:
 - resistance to plastic deformation or cracking in compression/Indentation.
 - better wear properties.



Hardness: Measurement

Table 6.5 Hardness Testing Techniques

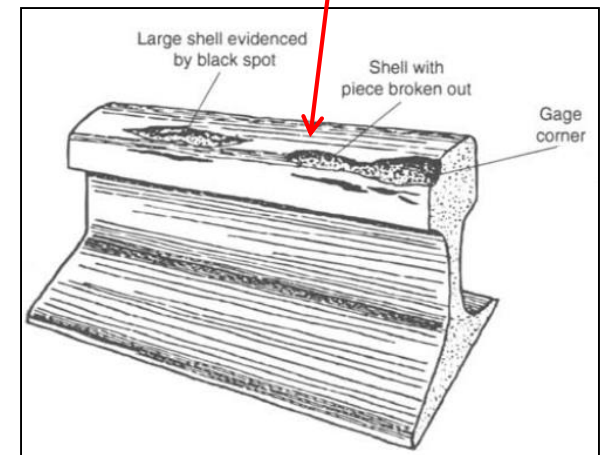
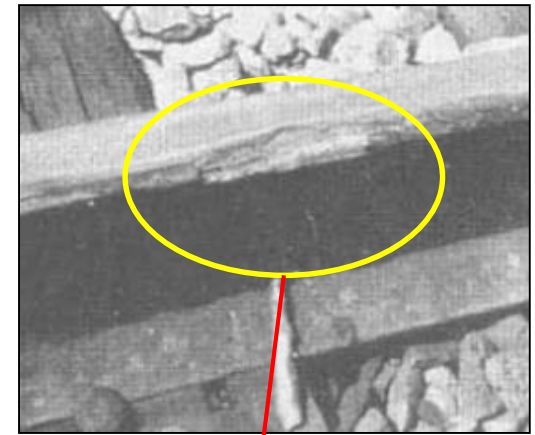
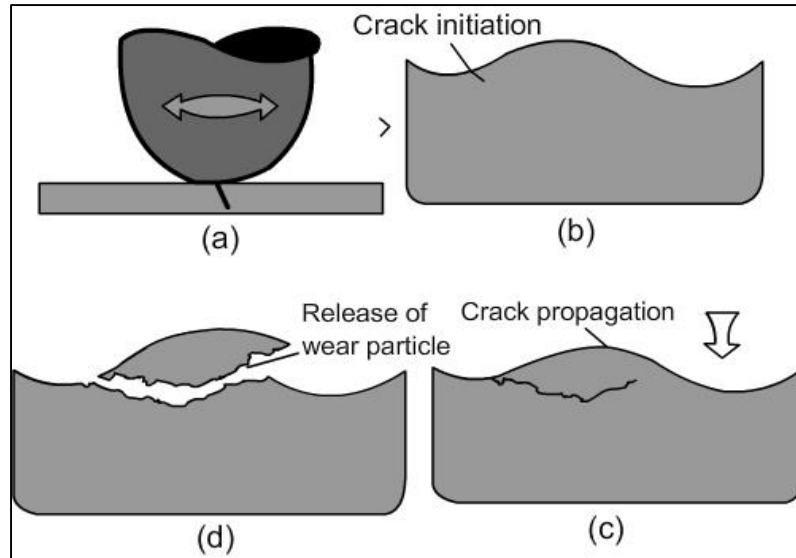
Test	Indenter	Shape of Indentation		Load	Formula for Hardness Number ^a
		Side View	Top View		
Brinell	10-mm sphere of steel or tungsten carbide			P	$HB = \frac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$
Vickers microhardness	Diamond pyramid			P	$HV = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			P	$HK = 14.2P/l^2$
Rockwell and Superficial Rockwell	Diamond cone $\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}$ in. diameter steel spheres	 	 	60 kg 100 kg 150 kg 15 kg 30 kg 45 kg	Rockwell Superficial Rockwell

^a For the hardness formulas given, P (the applied load) is in kg, while D , d , d_1 , and l are all in mm.

Source: Adapted from H. W. Hayden, W. G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*. Copyright © 1965 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

Surface fatigue

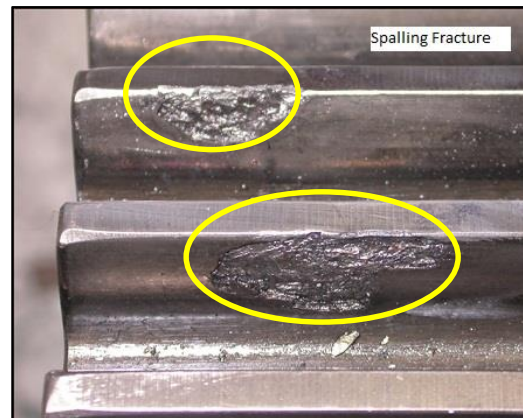
Two surfaces contacting to each other under pure rolling, or rolling with a small amount of sliding in contact



Rail road

Contact fatigue

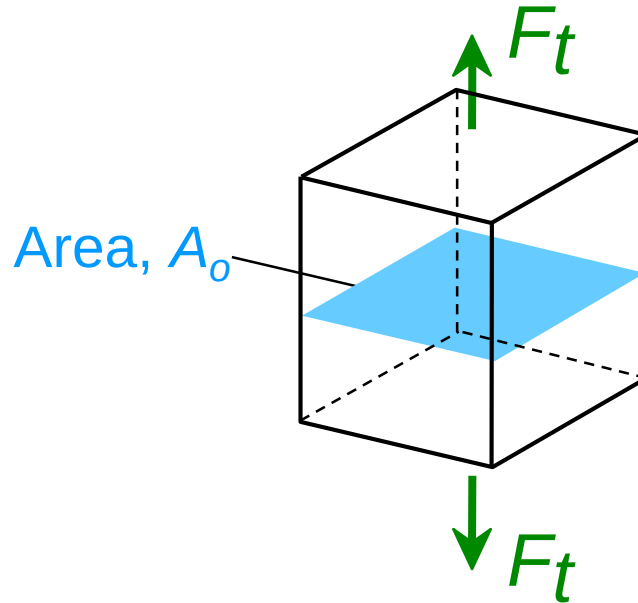
- As one element rolls many times over the other element
- Maximum shear stress is higher than fatigue limit



Metal spalling from gear tooth

Engineering Stress

Tensile stress, σ :

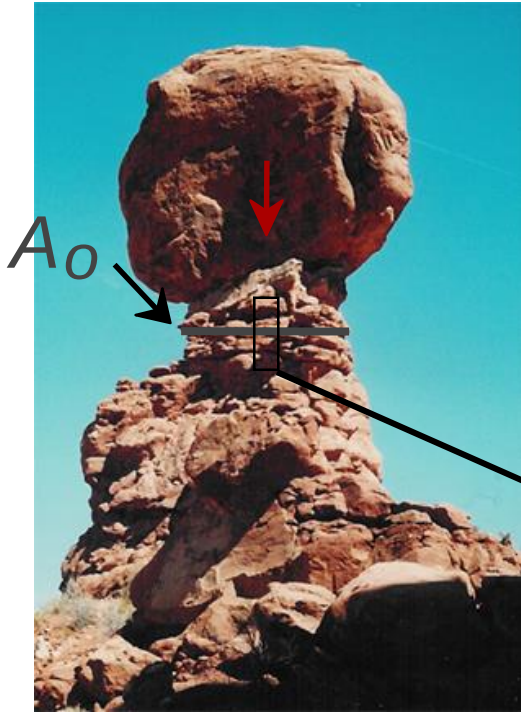


$$\sigma = \frac{F_t}{A_o} = \frac{\text{N}}{\text{m}^2}$$

original area
before loading

$\therefore$ Stress has units: N/m^2

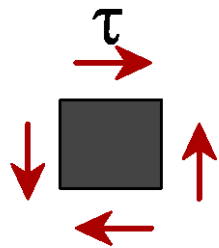
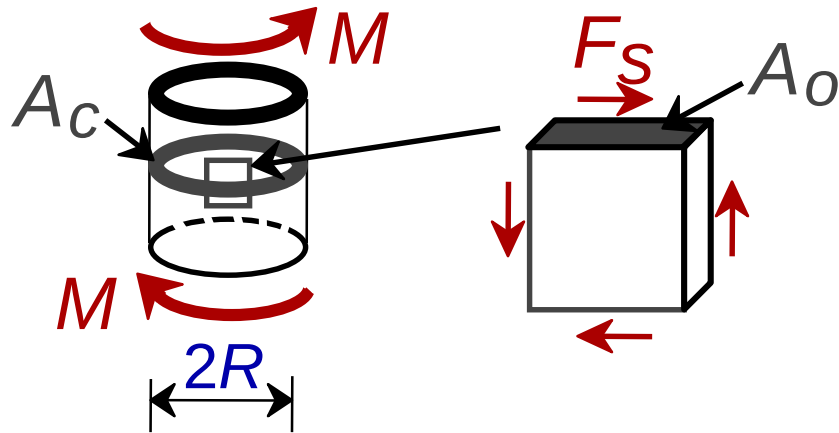
Simple compression:



$$\sigma = \frac{F}{A_o}$$

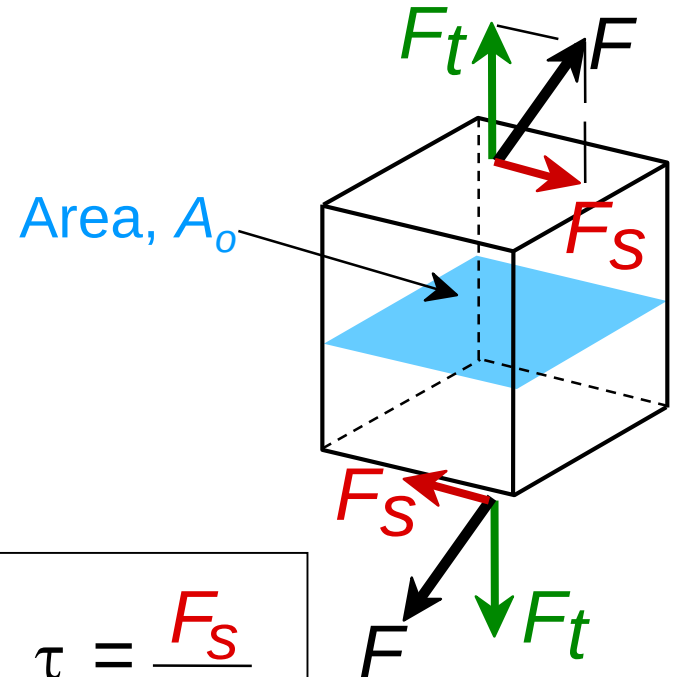


Torsion (a form of shear): drive shaft



$$\tau = \frac{F_s}{A_o}$$

- Shear stress, τ :



$$\tau = \frac{F_s}{A_o}$$

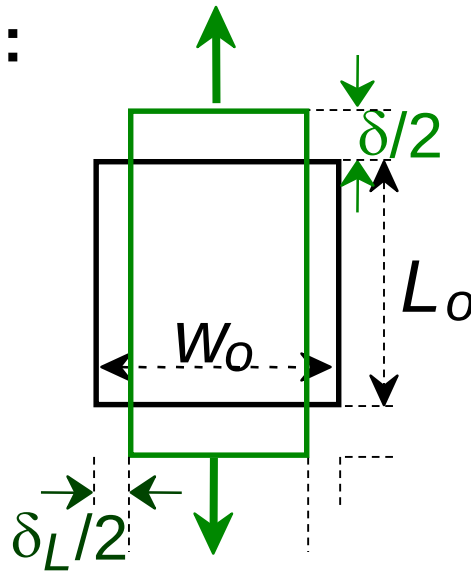
Engineering Strain

Tensile/Axial strain:

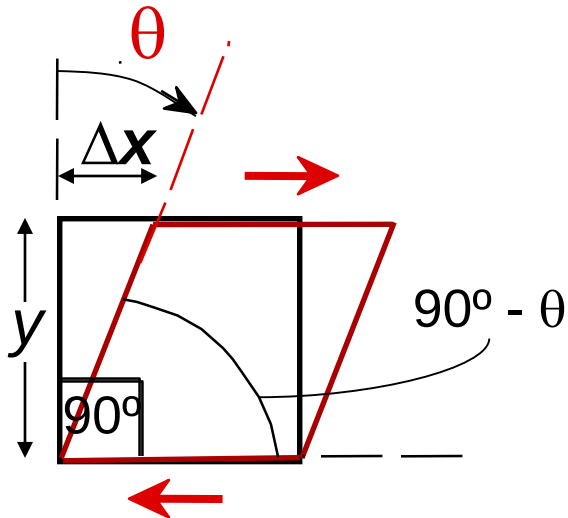
$$\varepsilon = \frac{\delta}{L_o}$$

Lateral strain:

$$\varepsilon_L = \frac{-\delta_L}{W_o}$$



Shear strain:



$$\gamma = \Delta x / y = \tan \theta$$

Strain is always dimensionless.

Linear Elastic Constant and its Relationships

Young's modulus or Modulus of elasticity (E) = Normal stress/ Normal strain

Shear modulus or Modulus of Rigidity (G) = Shear stress/ Shear strain

Bulk modulus (K) = Direct stress/ Volumetric strain

Poisson's ratio $\mu = -\frac{\varepsilon_T}{\varepsilon}$

ε = Longitudinal / Axial Strain

ε_T = Transverse / Lateral Strain

Linear Elastic Properties

Modulus of Elasticity or Young's modulus), E :

Modulus of elasticity is a **quantity** that measures an object resistance to being **deformed elastically** when a stress is applied to it.

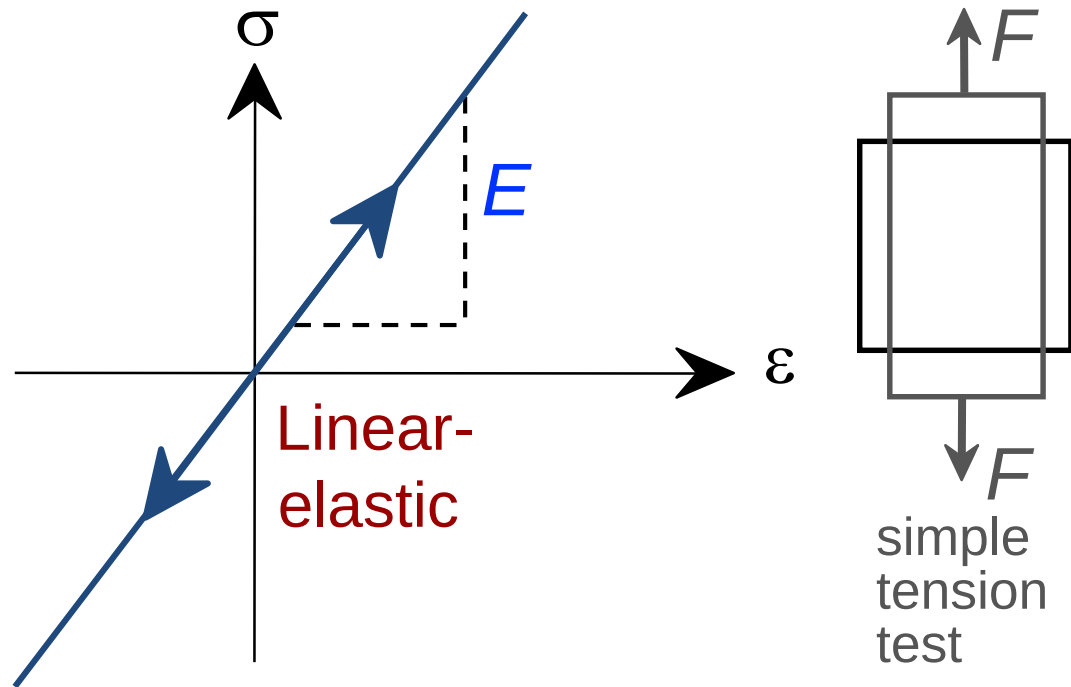
The elastic modulus of an object is defined as the slope of its stress–strain curve in the **elastic deformation region**.

Hooke's Law:

$$\sigma \propto \varepsilon$$

$$\sigma = E \varepsilon$$

$$E = \text{Stress/Strain}$$



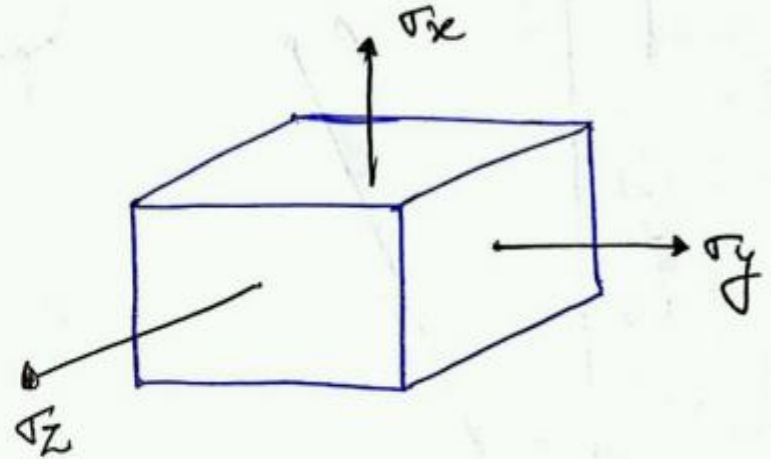
Hooke's law states that the **strain** of the material is proportional to the applied **stress** within the **elastic limit** of that material.

GENERALISED HOOKE'S LAW

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\mu \sigma_y}{E} - \frac{\mu \sigma_z}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\mu \sigma_x}{E} - \frac{\mu \sigma_z}{E}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\mu \sigma_x}{E} - \frac{\mu \sigma_y}{E}$$



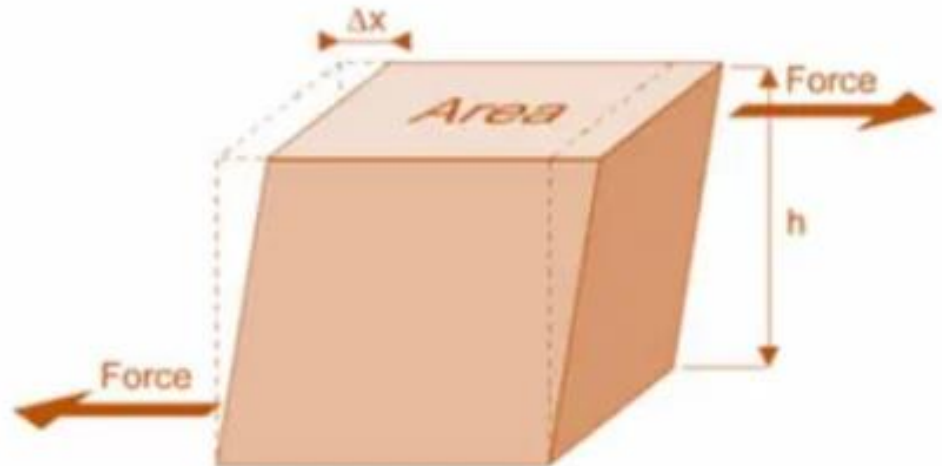
Shear modulus or Modulus of Rigidity (G)

When a body is subjected to shear stress the shape of the body gets changed, the ratio of shear stress to the corresponding shear strain is called rigidity modulus or modulus of rigidity.

It is denoted by the letters “G” or “C”.

Unit of rigidity modulus is MPa.

$$\text{Rigidity modulus}(G) = \frac{\text{Shear Stress}}{\text{Shear Strain}}$$



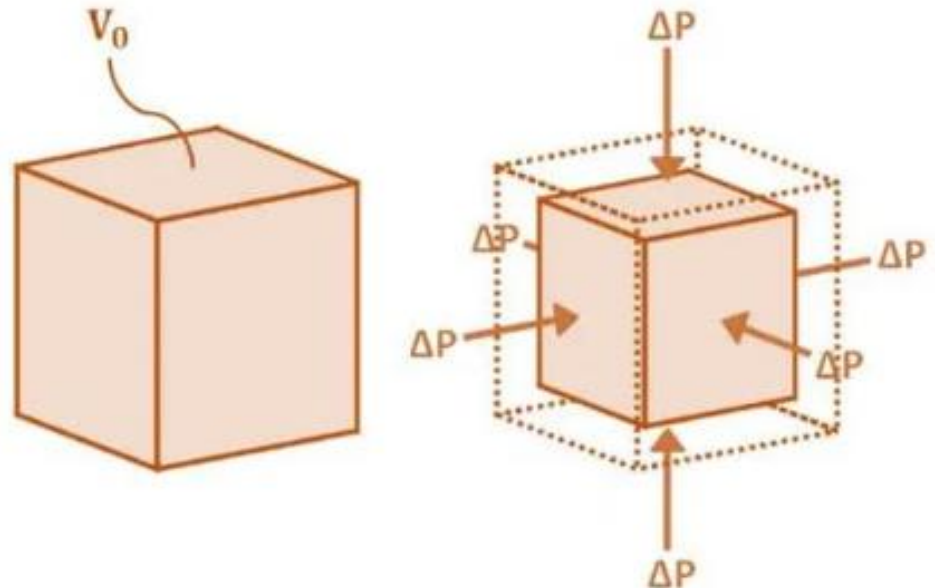
Bulk modulus (K) = Direct stress/ Volumetric strain

When a body is subjected to mutually perpendicular direct stresses which are alike and equal, within its elastic limits, the ratio of direct stress to the corresponding volumetric strain is found to be constant.

This ratio is called bulk modulus and is represented by letter “K”.

Unit of Bulk modulus is MPa.

$$\text{Bulk Modulus}(K) = \frac{\text{Direct Stress}}{\text{Volumetric Strain}}$$



Poisson's ratio

Poisson's ratio, ν : the expansion or contraction of a material in directions perpendicular to the direction of loading.

It is the ratio of transverse strain to Longitudinal strain.

$$\mu = -\frac{\varepsilon_T}{\varepsilon} = -\frac{\delta_L/W_o}{\delta/L_o}$$

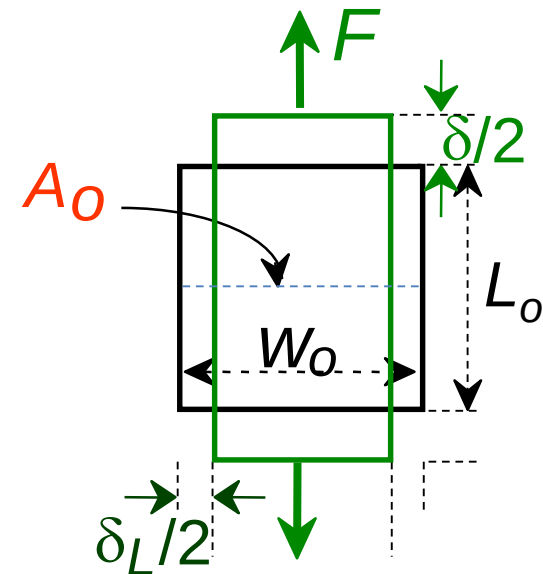
ε = Longitudinal / Axial Strain

ε_T = Transverse / Lateral Strain

metals: $\mu \sim 0.33$

ceramics: $\mu \sim 0.25$

polymers: $\mu \sim 0.40$



Units: μ : dimensionless

Linear Elastic Relationships

Simple tension:

$$\sigma = E \varepsilon$$

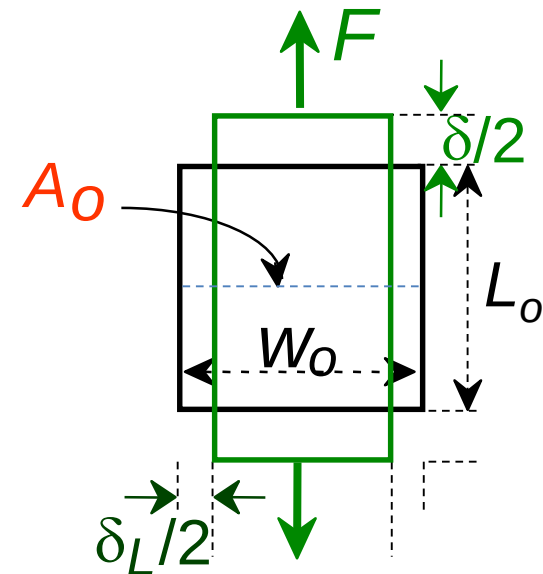
$$\delta = \frac{FL_o}{EA_o}$$

$$\delta_L = -\mu \frac{FW_o}{EA_o}$$

$$\mu = -\frac{\varepsilon_T}{\varepsilon} = -\frac{\delta_L/W_o}{\delta/L_o}$$

ε = Longitudinal / Axial Strain

ε_T = Transverse / Lateral Strain



Material, geometric, and loading parameters all contribute to deflection.
Larger elastic moduli minimize elastic deflection.

Linear Elastic Relationships

The relationship between Young's Modulus (E), rigidity Modulus (G) and Poisson's ratio (μ):

$$E = 2G(1 + \mu)$$

The relationship between Young's Modulus (E), Bulk Modulus (K) and Poisson's ratio (μ):

$$E = 3K(1 - 2\mu)$$

The relationship between Young's Modulus (E), Bulk Modulus (K) and rigidity Modulus (G):

$$E = \frac{9KG}{(3K + G)}$$

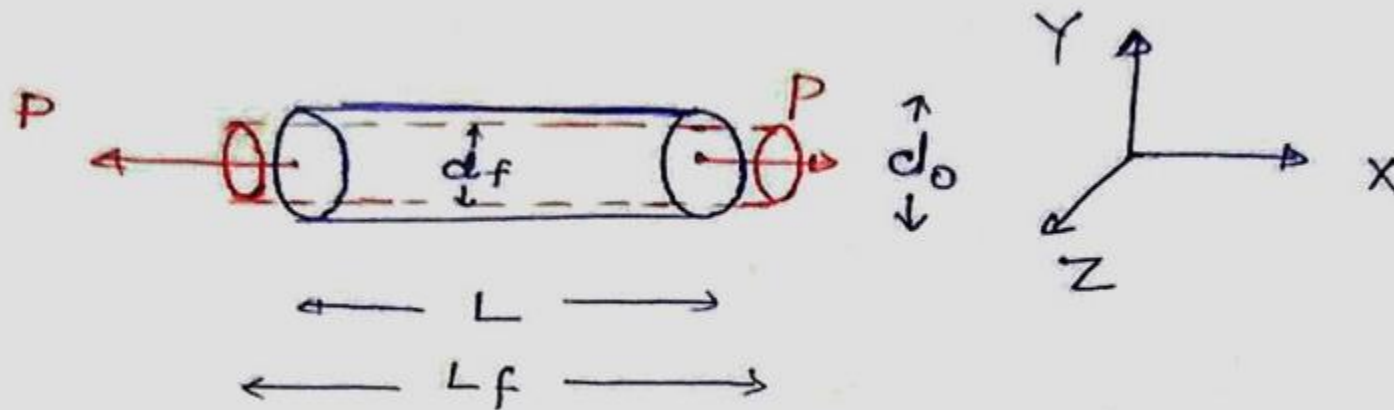
Linear Elastic Relationships

Poisson's ratio (μ) can be expressed in terms of Bulk Modulus (K) and rigidity Modulus (G) as:

$$\mu = \frac{(3K - 2G)}{(6K + 2G)}$$

A prismatic bar of diameter 30 mm is subjected to axial tensile load 5000 kN. It is found that the diameter is change to 29.891 mm. Determine Modulus of elasticity (E), Poisson's ratio (μ), change, if L = 1 m, Modulus of rigidity (G) = 70 GPa.

Linear Elastic Relationships



given $d_f = 29.891 \text{ mm}$, $d_o = 30 \text{ mm}$

$P = 5000 \text{ kN}$, $L = 1 \text{ m} = 1000 \text{ mm}$,

$G = 70 \text{ GPa}$

$E = ?$, $\mu = ?$, ~~600~~

$$E = 2G(1 + \mu) \quad \text{--- (1)}$$

Linear Elastic Relationships

$$\mu = ?$$
$$\mu = -\frac{\epsilon_y}{\epsilon_x} \text{ or } -\frac{\epsilon_z}{\epsilon_x} = \frac{-[d_f - d_o / d_o]}{[l_f - l / l]}$$

$$\mu = \frac{-[29.891 - 30 / 30]}{(l_f - 1000 / 1000)} \quad \text{--- (2)}$$

$$l_f = ?$$

$$\mu = -\frac{\epsilon_y}{\epsilon_x}$$

$$\epsilon_y = -\mu \epsilon_x = -\mu \frac{\sigma}{E}$$

$$\frac{d_f - d_o}{d_o} = -\mu \frac{\sigma}{E}$$

$$\frac{29.891 - 30}{30} = -\frac{\mu}{E} \times \frac{5000 \times 10^3}{\pi/4 (30)^2}$$

Linear Elastic Relationships

$$\frac{\mu}{E} = 5.183627878 \times 10^{-7} \quad \left(\frac{1}{\text{MPa}} \right)$$

from eqn ①

$$E = 2G(1 + \mu)$$

$$\frac{1}{2G} = \frac{1}{E} + \frac{\mu}{E}$$

$$\frac{1}{2 \times 70 \times 10^3} = \frac{1}{E} + 5.183627878 \times 10^{-7}$$

$$E = 150.954 \text{ GPa}$$

$$\therefore \frac{\mu}{E} = 5.183627878 \times 10^{-7}$$

$$\mu = \underline{0.07824}$$

Linear Elastic Relationships

Example 1.26 A bar, 24 mm in diameter and 400 mm in length, is acted upon by an axial load of 38 kN. The elongation of the bar and the change in diameter are measured as 0.165 mm and 0.0031 mm respectively. Determine

(i) Poisson's ratio, and (ii) the values of the three moduli

Solution

$$A = (\pi/4) 24^2 = 144 \pi \text{ mm}^2$$

$$\sigma = 38\,000 / 144 \pi = 84 \text{ MPa}$$

Lateral strain = ν . Linear strain

$$\frac{\delta d}{d} = \nu \frac{\delta L}{L} \text{ or } \frac{0.0031}{24} = \nu \frac{0.165}{400} \text{ or } \nu = 0.313$$

$$E = \frac{\sigma}{\epsilon} = \frac{84}{0.165/400} = 203\,636 \text{ MPa}$$

Also,

$$E = 2N(1 + \nu) = 3K(1 - 2\nu)$$

$\therefore$

$$G = \frac{E}{2(1 + \nu)} = \frac{203\,636}{2(1 + 0.313)} = 77\,546 \text{ MPa}$$

Linear Elastic Relationships

and

$$K = \frac{E}{3(1-2\nu)} = \frac{203\,636}{3(1-2 \times 0.313)} = 181\,494 \text{ MPa}$$

Linear Elastic Relationships

Example 1.27 A bar, 12 mm in diameter, is acted upon by an axial load of 20 kN. The change in diameter is measured as 0.003 mm. Determine
(i) Poisson's ratio and (ii) the modulus of elasticity and the bulk modulus.
The value of the modulus of rigidity is 80 GPa.

Solution

$$A = (\pi/4) 12^2 = 36 \pi \text{ mm}^2$$
$$\sigma = 20\,000 / 36 \pi = 176.84 \text{ MPa}$$

Lateral strain = ν . Linear strain

$$\frac{\delta d}{d} = \nu \epsilon \quad \text{or} \quad \frac{0.003}{12} = \nu \epsilon \quad \text{or} \quad \epsilon = 0.000\,25/\nu \quad (i)$$

Now, $E = 2G(1 + \nu) = 2 \times 80\,000(1 + \nu) = 160\,000 + 160\,000 \nu$

Also, $E = \frac{\sigma}{\epsilon} = \frac{176.84}{0.000\,25/\nu} = 707\,360 \nu$ [using (i)]

$$\therefore 707\,360 \nu = 160\,000 + 160\,000 \nu$$
$$\nu = 0.2923$$

Linear Elastic Relationships

Thus

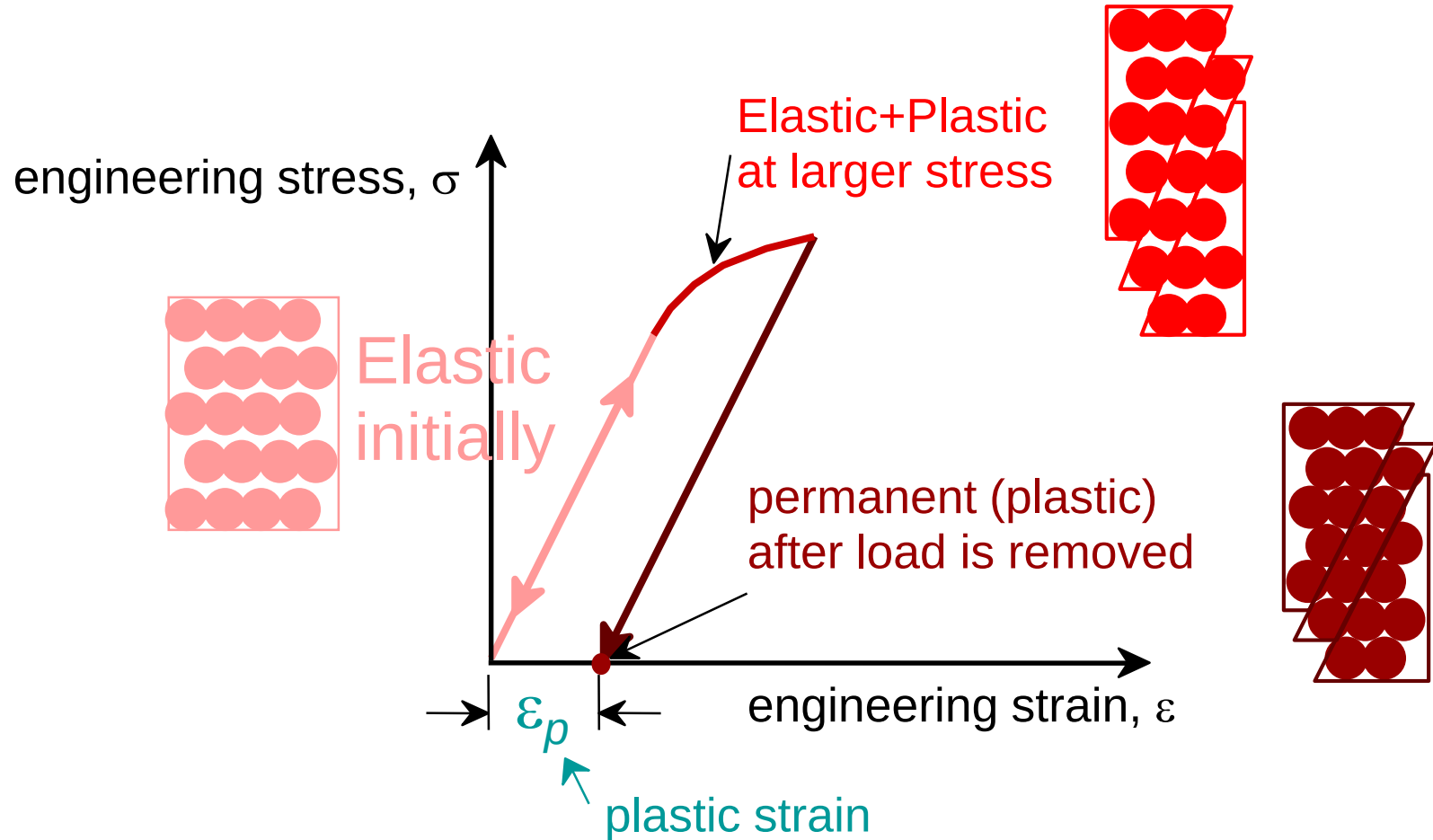
$$E = 707\,360 \times 0.2923 = 206\,771 \text{ MPa}$$

and

$$K = \frac{E}{3(1-2\nu)} = \frac{206\,771}{3(1-2 \times 0.2923)} = 165\,921 \text{ MPa}$$

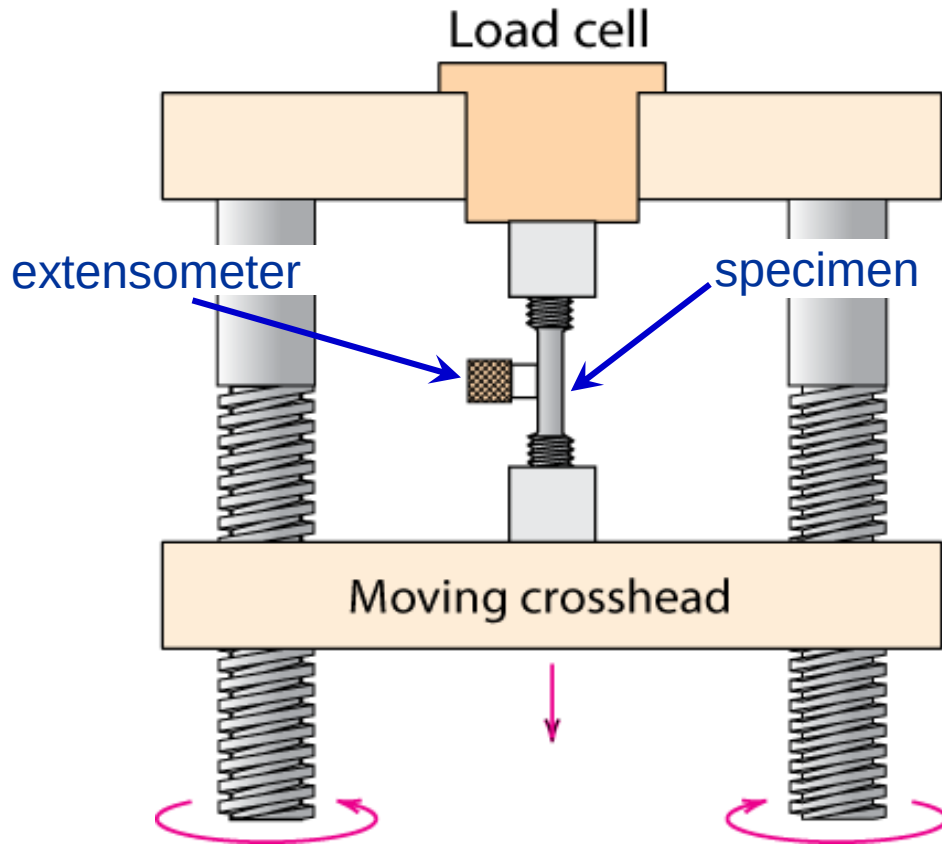
Plastic Deformation

Simple tension test:

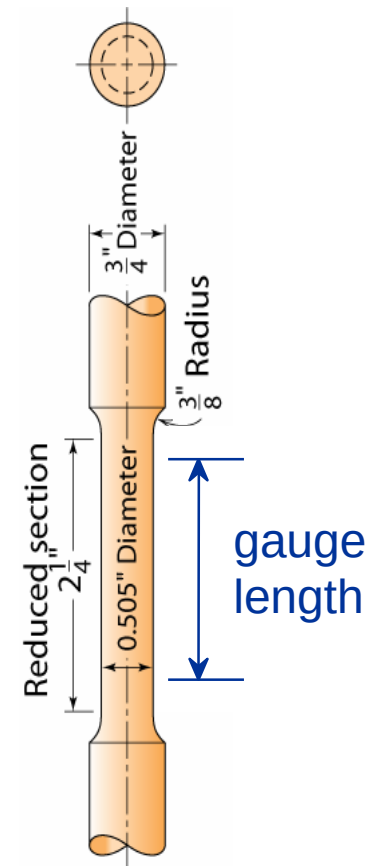


Stress-Strain Testing

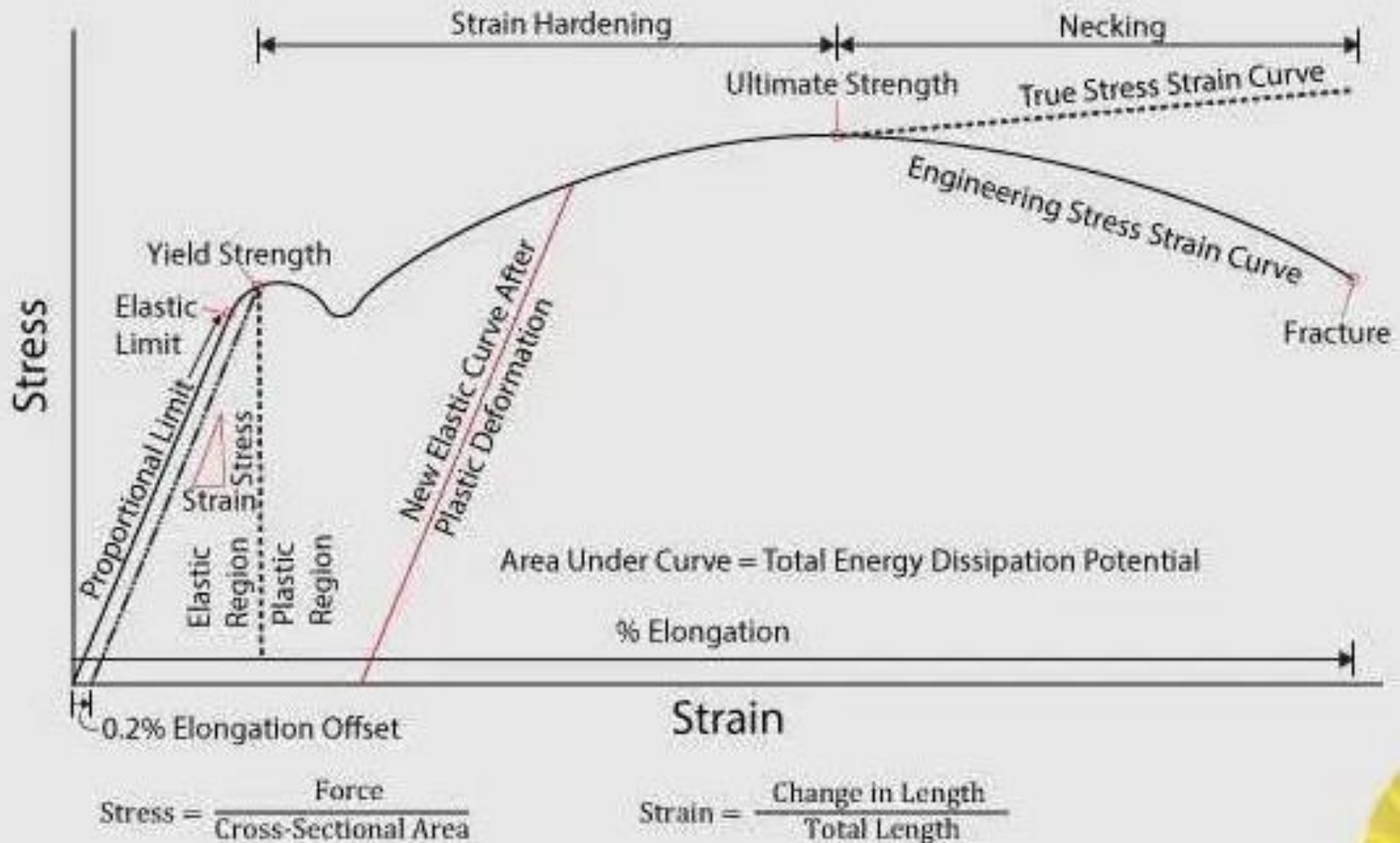
Tensile test machine



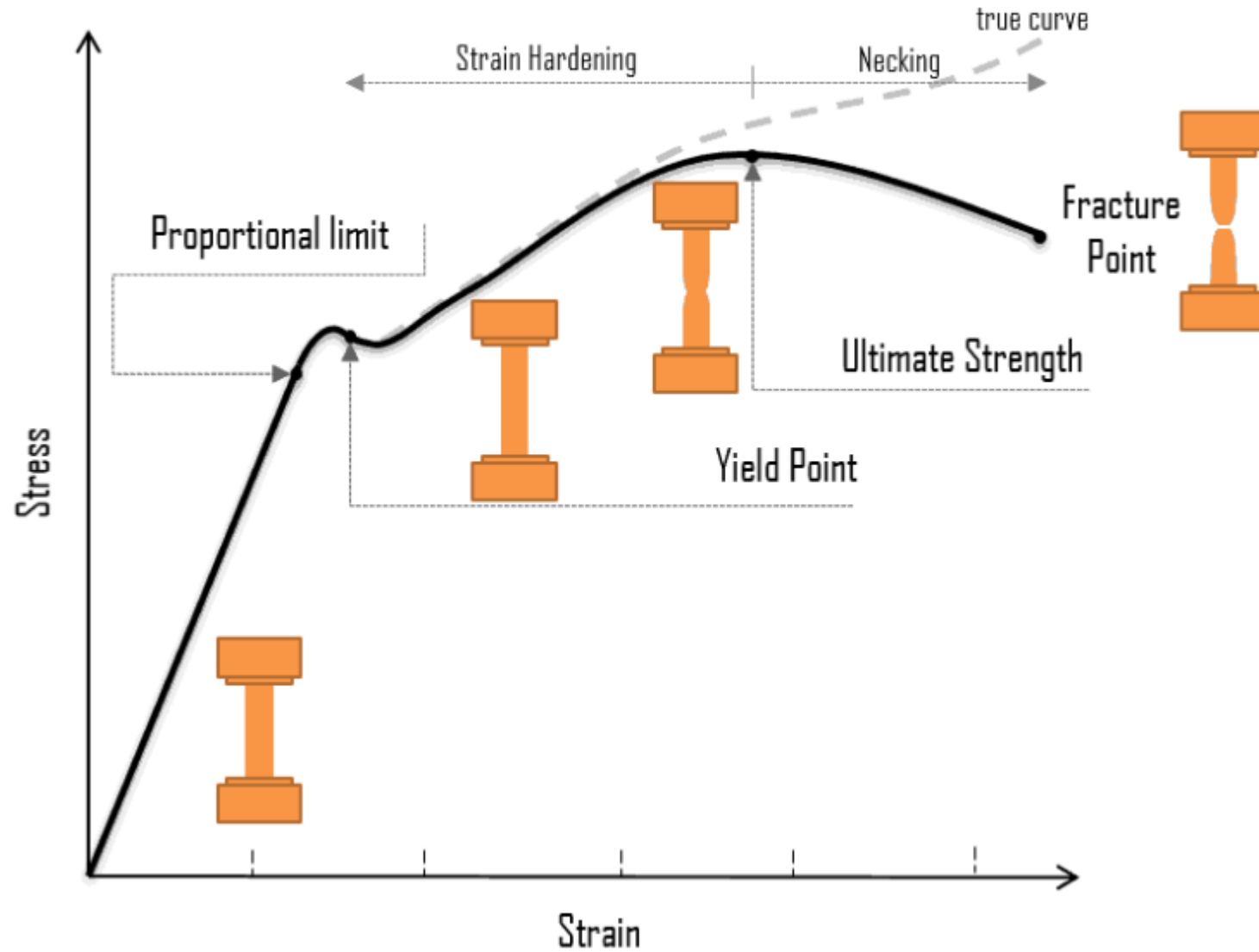
Tensile specimen



Stress & Strain Curve



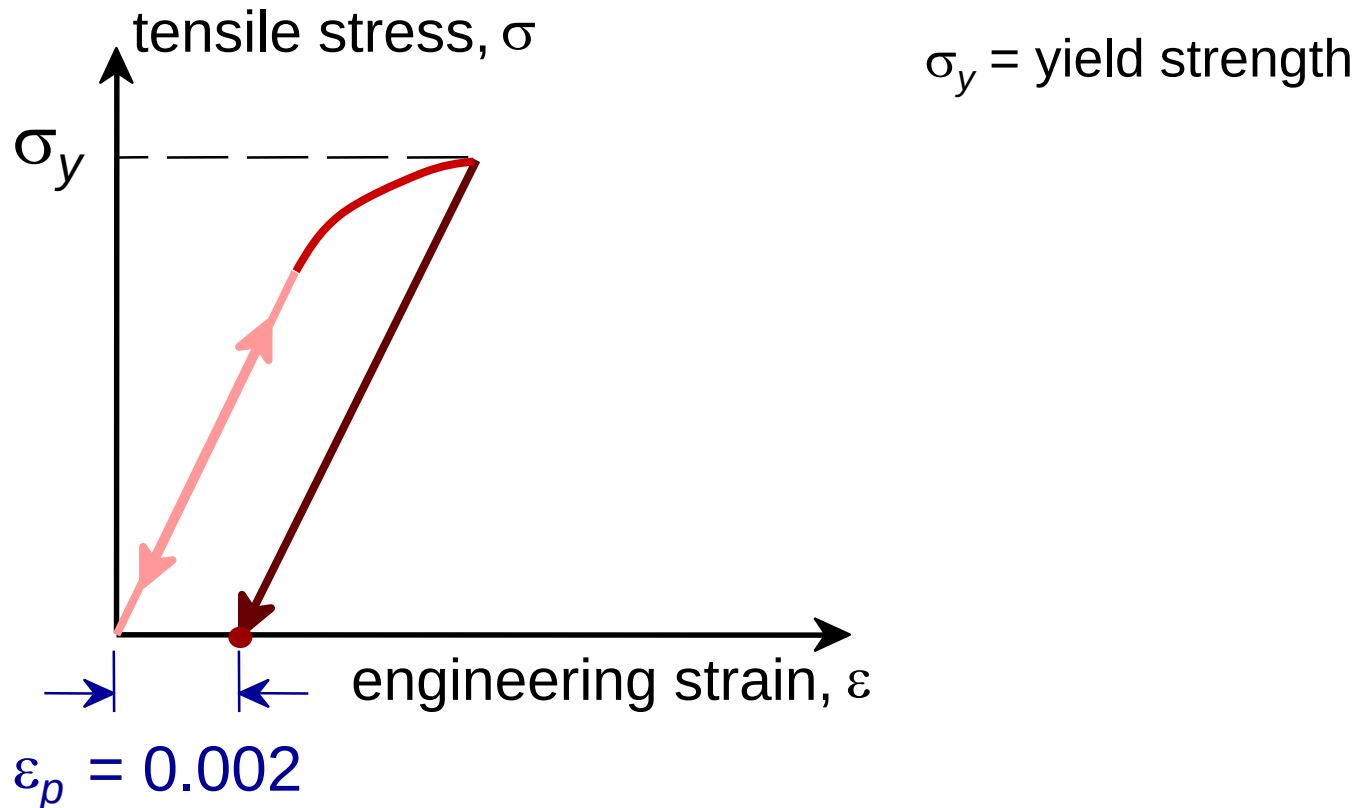
Stress & Strain Curve



Yield Strength

Stress at which *noticeable* plastic deformation has occurred.

when $\varepsilon_p = 0.002$



Thank You